NetTrace AI

Explainable fault localisation for ISP and enterprise networks

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Abstract

Problem

- Slow-internet complaints may stem from a faulty router, an overloaded link, a DNS failure or an upstream ISP issue.
- Engineers compare dashboards, alarms and logs by hand to tell causes from symptoms.
- Manual triage delays root-cause identification and extends downtime.
- Monitoring tools say that something is wrong, rarely why or where.

Approach

- Collect latency, loss, DNS, interface and route data from authorised sources.
- Detect anomalies against baselines, then correlate them with topology and event timing.
- Rank candidate routers and links using transparent scores and cited evidence.
- Validate on a simulated topology with injected faults and measured accuracy.

Background

Telemetry available

- Latency and packet loss from monitoring probes.
- DNS resolution results.
- Router and interface counters via SNMP [4].
- Traceroutes and route-change events [5].
- Network event logs.

Why diagnosis is hard

- One fault raises many alarms that describe symptoms, not the cause [1].
- Symptoms appear at points far from the fault.
- Healthy neighbours are needed as contrast.
- Correlation alone is not proof of causation.

Fault localisation: inferring the root cause of a failure from observed symptoms and network structure [1].

Literature & RFC Review

Source	Contribution	Use in NetTrace AI
Steinder & Sethi [1]	Survey of fault-localisation techniques	Frames our symptom-to-cause inference
Kompella et al. [2]	Risk modelling: shared components explain many symptoms	Scoring candidates by shared-path evidence
Duffield [3]	Infers link behaviour from end-to-end probes	Reasoning from multi-point probes to a link
RFC 3411 [4]	SNMP management architecture	Router and interface metrics
RFC 4271 [5]	BGP-4 protocol	Route changes as correlated evidence

Proposed Solution

1 Collect
Latency, loss, DNS results, interface metrics, traceroutes and event logs from authorised sources.

→

2 Detect
Compare measurements with baselines to flag rises in latency, loss, errors and route changes.

→

3 Correlate
Align timestamps, topology, affected probes and healthy neighbours to separate cause from symptom.

→

4 Localise
Score routers, links and services with a transparent method; list evidence for each rank.

→

5 Recommend
Output diagnosis, affected services and checks; verify recovery after the fix.

Design principles: explainable scores, no invented evidence, confidence reported only when calibrated on validated cases, correlation never treated as proof.

Optimization Proposed

Aspect	Before	With NetTrace AI
Visibility	Separate dashboards	Unified incident view
Alarms	Isolated, little context	Correlated evidence timeline
Diagnosis	Manual log comparison	Ranked candidate faults
Explanation	States what is wrong	Shows why a fault is suspected
Recovery	Checked across tools	Before-and-after verification

Illustrative simulated incident #1042

38% loss

Baseline: 2%

143 ms RTT

Baseline: 21 ms

R17 → upstream link

Suspected fault; adjacent links healthy

Solution Architecture & Design

Simulator
Normal traffic plus injected failures

→

Collector
Prometheus, SNMP-style metrics, ping and traceroute

→

Analysis engine
Python: anomalies, correlation, ranking

→

API layer
Node.js and Express

→

Dashboard
React, Tailwind, Recharts

Topology view
React Flow shows routers, links, probes and degraded paths.

Evidence store
PostgreSQL keeps telemetry, incidents and history.

Deployment
Docker packages every service for reproducible runs.

Figure: Layered architecture; no paid AI API is needed for the core system.

Data Flow Diagram

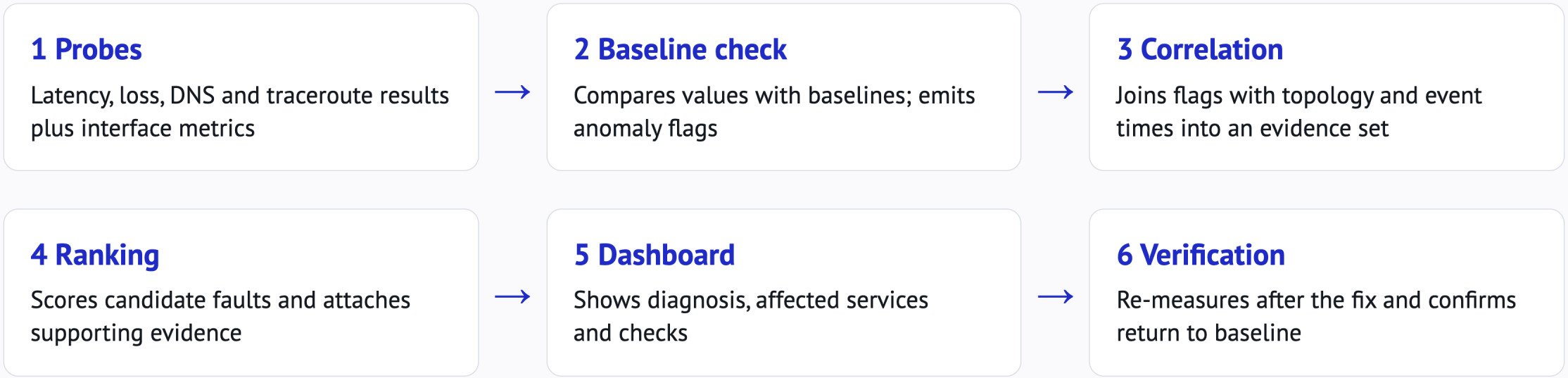


Figure: Data flow; verification results feed back into the baseline check.

Deployment Timeline

Phase	Milestone	Deliverable	Success check
1	Topology and simulator	Graph view; normal and faulty traffic	Baseline data is stable
2	Detection	Baselines and anomaly flags	Injected faults are flagged
3	Localisation	Scored ranking with evidence	Top-1 matches injected fault
4	Dashboard	Incidents, history, recovery view	Metrics return to normal after fix
5	Evaluation and pilot	Repeatable fault-injection runs	Accuracy, delay, false positives measured

Metrics: top-1 localisation accuracy, detection delay, false-positive rate, time to root cause.

Bibliography

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[5] Y. Rekhter et al., RFC 4271: A Border Gateway Protocol 4 (BGP-4), IETF, 2006.